

INF1004 Mathematics II

Tutorial 5: Singular Value Decomposition (Modified)

SIT

Question 1: Orthonormal Matrices

Show that the following matrices are orthonormal.

Recall: A matrix Q is orthonormal if $Q^T Q = I$ (or equivalently $Q^{-1} = Q^T$).

$$A = \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}$$

Question 2: SVD of a Rectangular Matrix

Compute the Singular Value Decomposition (SVD) of the matrix:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Hint: Start by computing AA^T to find the singular values σ and the left singular vectors U .

Question 3: SVD of a Symmetric Matrix

Compute the Singular Value Decomposition of the matrix:

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

Note: For symmetric positive-definite matrices, the Singular Values are related directly to the Eigenvalues.

Question 4: SVD of a Singular Matrix

Compute the Singular Value Decomposition of the matrix:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Hint: Since the rows are identical, this matrix has a determinant of 0. Expect one of your singular values to be 0.

Question 5: SVD of a 3x4 Matrix

Compute the Singular Value Decomposition ($A = U\Sigma V^T$) of the following 3×4 matrix:

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Tips for calculation:

- Step 1: Compute AA^T (which is 3×3) to find U and Σ .
- Step 2: Notice that Row 1 and Row 3 are identical. What does this tell you about the matrix?
- Step 3: Use the formula $v_i = \frac{1}{\sigma_i} A^T u_i$ to find the columns of V .